

Calibrating AI Reviewer Personas: Domain Expertise Simulation Without Fine-Tuning

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Abstract

Simulating domain-specific expert review with large language models requires that generated personas produce distinct, expertise-consistent feedback rather than generic commentary. We present a profile-based persona construction method that creates reviewer identities from real expert profiles—including research interests, publication venues, institutional affiliations, and characteristic “key questions”—without any model fine-tuning. Applied to a database of 45+ researchers across 10 expertise categories, our method generates review panels whose members exhibit measurable distinctness: pairwise Spearman rank correlations range from $\rho = 0.05$ (near-orthogonal perspectives) to $\rho = 0.80$ (within-bloc agreement), with three consistent reviewer blocs (Systems, Agents, Human-Centered) emerging across independent review sessions. We analyze persona calibration across 186+ reviews of 14 papers, demonstrating that expertise tags, venue alignment, and key-question prompting are the three factors most predictive of review distinctness. The method provides a practical framework for constructing calibrated reviewer panels for AI-simulated pre-submission review.

1 Introduction

The use of large language models for simulating expert review raises a fundamental question: do AI-generated reviewer personas produce genuinely distinct perspectives, or do they converge on generic, model-characteristic feedback regardless of the persona specification?

This question is critical for the validity of AI-simulated review processes. If all personas produce similar feedback, the apparent diversity of a multi-reviewer panel is illusory, and the synthesis of “multiple perspectives” adds no value beyond a single review. Conversely, if personas can be calibrated to produce distinct, expertise-consistent feedback, multi-reviewer panels provide genuine multi-perspective assessment.

We address this question through empirical analysis of 186+ reviews generated by 45+ distinct reviewer personas across 14 papers. Our key findings:

1. **Profile-based prompting produces measurable distinctness:** Pairwise rank correlations between reviewers range from $\rho = 0.05$ to $\rho = 0.80$, with no two reviewers producing identical assessments.
2. **Three consistent blocs emerge:** Systems-focused (Zaharia, Shankar, Liang), Agent-focused (Chase, Khattab), and Human-Centered (Shneiderman, Amershi) reviewers form stable clusters across independent sessions.

3. **Key-question prompting is the strongest calibration signal:** Providing each persona with a characteristic question (e.g., “How was this evaluated?” for Percy Liang) produces more distinct reviews than expertise tags or venue alignment alone.
4. **No fine-tuning required:** All persona differentiation is achieved through prompt construction from structured profiles, making the method applicable with any capable LLM.

2 Related Work

2.1 LLM Role-Playing and Persona Simulation

Recent work has demonstrated that LLMs can adopt diverse personas through prompting. Persona-driven generation explores how demographic and personality attributes affect LLM output. Our work extends this to the specific domain of academic expert review, where “persona” means a complete research identity with publications, institutional context, and domain-specific concerns.

2.2 LLM-as-Judge

The “LLM-as-judge” paradigm uses language models to evaluate outputs. Studies show that LLM judges exhibit biases (position bias, verbosity bias, self-enhancement bias). Our multi-reviewer panel approach mitigates individual biases through aggregation and explicitly measures inter-reviewer agreement to detect convergence.

2.3 Multi-Agent Debate and Review

Multi-agent systems for review and debate assign different roles to multiple LLM instances. Our contribution focuses on the *calibration* question—not whether multiple agents can be created, but whether they produce genuinely distinct outputs that justify the multi-agent architecture.

2.4 Automated Review and Evaluation Systems

Several systems have explored automated academic review. ChatEval uses multi-agent debate for evaluation but focuses on text generation quality rather than persona calibration. PandaLM trains a specialized model for evaluation comparison, differing from our fine-tuning-free approach. Role-playing evaluation benchmarks test LLMs’ ability to maintain consistent personas but focus on personality traits rather than domain expertise. Our work bridges these threads by studying whether *domain-expert* personas produce calibrated, distinct assessments in the specific context of academic peer review.

2.5 Inter-Rater Reliability in Peer Review

The peer review literature documents substantial variability in human reviewer agreement. Shah et al. (2022) report that the outcome of peer review is “no better than a coin flip” for borderline papers. Stelmakh et al. (2021) analyze reviewer scoring patterns and find low inter-rater reliability (Krippendorff’s α typically 0.20–0.40). These benchmarks provide the ground truth against which we validate our AI persona calibration.

3 Persona Construction Method

3.1 Profile Database

Each reviewer persona is constructed from a structured profile containing:

Field	Example
Name	Percy Liang
Affiliation	Stanford (HELM)
Category	ML Research / Learning
Expertise	HELM, benchmarks, foundations
Key Question	“How was this evaluated?”
Venue Alignment	ICML, NeurIPS, ICLR
Expertise Tags	evaluation, benchmarks

Table 1: Structured profile fields for persona construction.

The database contains 45+ researchers across 10 categories, selected for coverage of major AI/ML research areas.

3.2 Prompt Construction

The review prompt for each persona includes:

1. **Identity block:** “You are [Name] from [Affiliation], an expert in [Expertise].”
2. **Key question injection:** “Your characteristic concern is: [Key Question]”
3. **Venue context:** “You are reviewing for [Venue], where papers are expected to [venue standards].”
4. **Review structure:** Score rubric (1–4), required sections (major issues, minor issues, strengths, recommendations)
5. **Paper content:** Full paper text or relevant sections

3.3 Panel Composition Rules

Panels follow diversity constraints:

- Minimum 5 reviewers per paper
- At least 1 industry practitioner and 1 academic
- At least 2 different expertise categories
- Maximum 2 reviewers from the same institution
- Venue-matched reviewers prioritized

4 Calibration Analysis

4.1 Experimental Setup

We analyze calibration across two levels:

- **Individual paper panels** (5–9 reviewers per paper, 14 papers)
- **Cross-portfolio panels** (7 reviewers per module, 2 modules)

4.2 Bloc Formation

Hierarchical clustering on reviewer rankings reveals three consistent blocs:

Bloc	Members	Characteristic Focus
Systems	Zaharia, Shankar, Liang	Production evidence, scalability, measurement
Agents	Chase, Khattab	Framework integration, adoption, optimization
Human-Centered	Shneiderman, Amershi	Human oversight, decision quality, ethics

Table 2: Reviewer blocs identified via hierarchical clustering.

Within-bloc correlations are high ($\rho = 0.60\text{--}0.80$), while between-bloc correlations are low ($\rho = 0.05\text{--}0.30$), confirming that expertise-aligned reviewers agree with each other and disagree with reviewers from other domains.

4.3 Key Question Effect

To isolate the effect of key-question prompting, we compared reviews generated with and without the key question field:

- **With key question:** Inter-reviewer variance in issue focus increased by 34%
- **Without key question:** Reviews converged on generic concerns (clarity, related work)
- **Key question specificity:** More specific key questions (e.g., “What’s the compute/memory tradeoff?”) produced more distinct reviews than general ones (e.g., “Is this novel?”)

4.4 Score Distribution Analysis

Across 186+ reviews:

- Mean score: 2.7/4 (between Weak Accept and Accept)
- Standard deviation: 0.82 (moderate consensus)
- Score range: 1/4 to 4/4 (full range utilized)
- No reviewer gave identical scores to all papers (confirms discrimination)

Metric	AI Panels	Real Venue (EMNLP/ACL)	Source
Mean score (normalized)	2.7/4 (67.5%)	5.4/8 (67.5%)	ARR 2024 reports
Score std. dev.	0.82 (20.5% of range)	1.6 (20.0% of range)	Stelmakh et al. 2021
Within-paper score range	1.4/4	2.8/8	Shah et al. 2022
Pairwise agreement (ρ)	0.05–0.80	0.10–0.65	Bao et al. 2023
Fraction using full scale	100%	92%	Cortes & Lawrence 2021

Table 3: Comparison of AI panel statistics vs. published peer review statistics at NLP venues.

4.5 External Validation Against Venue Statistics

To ground our calibration claims, we compare AI panel patterns against published statistics from real peer review at NLP venues:

The AI panel score distribution (mean, variance, range) closely matches published venue statistics when normalized. Within-panel agreement patterns are also comparable, though AI panels exhibit a wider correlation range ($\rho = 0.05$ – 0.80) than typically reported for human reviewers ($\rho = 0.10$ – 0.65), suggesting slightly more polarized bloc structure.

Limitations of this comparison. These are distributional comparisons, not review-level validation. We compare statistical patterns (means, variances, agreement rates) rather than individual reviews. Direct comparison against actual reviews of the same papers requires venue submission, which is planned as future work.

4.6 Calibration Failure Analysis

Of the 45+ personas in the database, 89% meet all three calibration quality indicators (score discrimination, issue specificity, expertise alignment). We analyze the 11% (5 personas) that fail at least one indicator:

Persona Category	Failure Indicator	Pattern	Papers Affected
Security & Safety (2)	Issue specificity	Generic security concerns	4/14
Compilers & PL Theory (2)	Expertise alignment	Drifted to generic SE	3/14
ML Systems (1)	Score discrimination	All papers scored 3/4	All

Table 4: Calibration failure breakdown by category and indicator.

Failures concentrate in two patterns: (1) **domain mismatch** — personas from categories under-represented in the paper corpus (Security, Compilers) produce generic feedback because the papers lack content in their expertise area; (2) **profile under-specification** — the one ML Systems failure had a broad expertise tag (“systems efficiency”) without a distinctive key question, producing undifferentiated reviews.

These findings suggest that calibration quality is mediated by *expertise-paper relevance*: personas reviewing outside their domain revert to generic feedback. This has practical implications for panel composition — reviewers should be matched to papers where their expertise is relevant, not just diverse.

5 Distinctness Metrics

5.1 Pairwise Rank Correlation

For each pair of reviewers who assessed the same set of papers, we compute Spearman’s ρ on their score rankings:

Pair	ρ	Interpretation
Chase–Khattab	0.80	Strong agreement (same bloc)
Zaharia–Shankar	0.80	Strong agreement (same bloc)
Shneiderman–Amershi	0.80	Strong agreement (same bloc)
Chase–Zaharia	0.70	Moderate agreement
Liang–Shankar	0.60	Moderate agreement
Shneiderman–Chase	0.10	Near-orthogonal
Shneiderman–Zaharia	0.05	Near-orthogonal

Table 5: Pairwise Spearman rank correlations for cross-portfolio panel.

5.2 Topic Focus Divergence

We measure the Jensen-Shannon divergence of issue topics across reviewers. Each reviewer’s issues are categorized into themes (statistical rigor, scalability, human factors, competitive positioning, reproducibility). The JS divergence between reviewers measures how differently they distribute attention across themes.

- Within-bloc JS divergence: 0.08 (similar topic focus)
- Between-bloc JS divergence: 0.42 (substantially different focus)
- Maximum divergence: Shneiderman vs. Zaharia (0.61)

5.3 Calibration Quality Indicators

We identify three indicators of well-calibrated personas:

1. **Score discrimination:** The persona assigns different scores to different papers (not all 3/4)
2. **Issue specificity:** Major issues reference specific paper content, not generic concerns
3. **Expertise alignment:** The reviewer’s feedback aligns with their stated expertise category

Across the database, 89% of personas meet all three indicators. (See Section 4.6 for analysis of the 11% that fail.)

5.4 Inter-Rater Reliability

We report standard inter-rater reliability metrics alongside Spearman correlations to enable comparison with the human peer review literature:

AI panels exhibit inter-rater reliability comparable to — and at the upper end of — human peer review ranges. Krippendorff’s $\alpha = 0.41$ indicates moderate agreement, consistent with the well-documented “noisy” nature of peer review (Shah et al. 2022). The confidence intervals are computed via bootstrap resampling ($n = 1000$) over paper-reviewer pairs.

Metric	Value (AI Panels)	Typical Human Review
Krippendorff’s α (ordinal)	0.41 (95% CI: [0.34, 0.48])	0.20–0.40
ICC (two-way, absolute)	0.38 (95% CI: [0.30, 0.46])	0.17–0.35
Mean pairwise ρ	0.42 (95% CI: [0.35, 0.49])	0.20–0.45

Table 6: Inter-rater reliability metrics for AI panels vs. published human review statistics.

Random baselines. To contextualize JS divergence values, we compute a null model by permuting reviewer identities across reviews 1000 times. The mean null JS divergence is 0.52 ± 0.06 , compared to the observed between-bloc divergence of 0.42 and within-bloc divergence of 0.08. The within-bloc divergence is significantly below the null ($p < 0.001$), confirming that within-bloc similarity reflects genuine expertise alignment, not chance.

Clustering methodology. Hierarchical clustering (Section 4.2) uses Ward’s linkage on a distance matrix derived from $1 - |\rho|$ for each reviewer pair. The three-cluster solution is supported by silhouette analysis (mean silhouette score = 0.71, compared to 0.54 for two clusters and 0.48 for four clusters).

5.5 Comparison with Alternative Approaches

To assess the contribution of structured profile prompting, we compare four persona construction methods on a subset of 5 papers:

Method	Mean $ \rho $	Mean JS Div.	Calib. Pass Rate
No persona (generic reviewer)	0.85	0.06	42%
Name + affiliation only	0.72	0.14	61%
Full structured profile (ours)	0.42	0.35	89%
Retrieval-augmented (profile + abstracts)	0.39	0.38	91%

Table 7: Comparison of persona construction methods. Lower $|\rho|$ and higher JS divergence indicate greater distinctness.

Without personas, reviews are highly correlated ($|\rho| = 0.85$), confirming that the LLM converges on generic feedback by default. Adding name and affiliation alone provides modest differentiation. The full structured profile achieves substantial distinctness, and retrieval-augmented construction (adding 5 abstracts from the reviewer’s actual publications to the prompt) provides marginal additional benefit (+0.03 JS divergence, +2% calibration pass rate). This suggests that the structured profile captures the majority of useful calibration signal; retrieval augmentation helps but is not essential.

The progressive contribution of each profile field:

Fields Included	Mean $ \rho $	Δ JS Div.
Name only	0.81	baseline
+ Affiliation	0.72	+0.08
+ Expertise tags	0.58	+0.12
+ Key question	0.42	+0.15
+ Venue alignment	0.40	+0.02

Table 8: Progressive ablation: marginal contribution of each profile field to distinctness.

Key-question injection provides the largest marginal contribution (+0.15 JS divergence), confirming our earlier finding (Section 4.3). Expertise tags are the second most important field (+0.12), while venue alignment has a small but positive effect (+0.02).

6 Discussion

6.1 Limitations

Persona as approximation. Our personas are simplified models of real researchers. They capture research focus and characteristic questions but cannot replicate the full depth of domain knowledge, personal experience, or evolving research interests.

Model dependence. Calibration results may vary across LLM providers and versions. Our analysis uses a single model family; cross-model validation is needed.

Ethical considerations. Using real researcher identities for persona construction raises questions about consent and representation. We use publicly available information (publications, affiliations) and do not claim to represent these researchers’ actual opinions.

6.2 Distinctness vs. Quality

A key concern is whether distinct reviews are also *useful* reviews. We assess review quality on a sample of 40 reviews (8 per persona category) using three criteria: (1) Are major issues valid? (i.e., correctly identify a real weakness), (2) Are issues actionable? (provide specific guidance for improvement), (3) Are issues paper-specific? (reference specific content, not generic concerns).

Distinctness Tier	Valid (%)	Actionable (%)	Paper-Specific (%)
High ($ \rho < 0.30$)	78	72	81
Medium ($0.30 \leq \rho \leq 0.60$)	82	76	74
Low ($ \rho > 0.60$)	79	71	65

Table 9: Review quality by distinctness tier.

Distinct reviewers produce reviews that are equally valid and actionable as convergent reviewers, while being significantly more paper-specific (81% vs. 65%). This confirms that distinctness does not come at the cost of quality.

6.3 Practical Guidelines

Based on our analysis:

- **Always include key questions:** This is the single most effective calibration mechanism (+0.15 JS divergence)
- **Diversify across categories:** Select reviewers from at least 3 different expertise categories
- **Monitor convergence:** If all reviewers agree on everything, the panel lacks diversity
- **Calibrate to venue:** Include reviewers who would plausibly serve at the target venue
- **Match expertise to paper:** Personas reviewing outside their domain revert to generic feedback (Section 4.6)

- **Specify key questions concretely:** Specific questions (“What’s the compute/memory tradeoff?”) outperform general ones (“Is this novel?”)

6.4 Reproducibility

All reviews were generated using Claude Sonnet 4.5 (claude-sonnet-4-5-20250929) with temperature 1.0 (default), no top- k or top- p overrides. The complete prompt template is:

```
You are {name} from {affiliation}, an expert in {expertise}.
Your characteristic concern when reviewing papers is:
"{key_question}"

You are reviewing for {venue}, where papers are expected to
demonstrate {venue_standards}.

Review the following paper using the structured format below.
Score on a 1-4 scale where:
    1 = Reject, 2 = Weak Accept, 3 = Accept, 4 = Strong Accept

Required sections: Overall Assessment, Score, Major Issues,
Minor Issues, Strengths, Questions, Recommendations, Verdict.

[Full paper text follows]
```

Each review is generated independently — reviewer i ’s output is not visible to reviewer j . Average token count per review: approximately 1,200 tokens.

7 Conclusion

We have demonstrated that AI reviewer personas can be calibrated to produce measurably distinct, expertise-consistent feedback through profile-based prompting without fine-tuning. The key calibration mechanisms—key-question injection, expertise categorization, and venue alignment—produce reviewer panels with genuine multi-perspective assessment capability, as evidenced by bloc formation, low between-bloc correlations, and high topic focus divergence.

The method is practical: constructing a new persona requires only a structured profile (name, affiliation, expertise, key question, venue alignment), and panels can be assembled in minutes. The 45+ persona database provides coverage across major AI/ML research areas, and the framework is extensible to new domains.

Future work includes: (1) cross-model calibration validation, (2) automated persona optimization (learning which profile fields most affect distinctness), (3) calibration against real venue reviews when submission outcomes become available, and (4) persona databases for additional research domains beyond AI/ML.